

Solution Requirements

Date	15 March 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system allows users to securely access the Smart Lender web application and submit loan application details for prediction.	High
2	Authorization levels	Loan Applicants can submit applications and view predictions, while Bank Administrators can manage the trained model, dataset, and application settings.	High
3	External interfaces	The system interacts with the Machine Learning model (model.pkl), Scaler (scaler.pkl), Loan Dataset.	High

4	Transactions processing	The system receives applicant information, validates inputs, preprocesses data, performs loan eligibility prediction, and returns the approval or rejection result.	Medium
5	Reporting	The system displays the loan prediction result, confidence score, and recommendation through the web interface.	High
6	Business rules, etc.	Loan eligibility is determined based on applicant income, co-applicant income, loan amount, loan term, credit history, education, employment status, dependents, and property area using the trained machine learning model.	Medium
7	Compliance to laws or regulations	The system protects applicant information, ensures data confidentiality, and follows applicable banking and data privacy practices.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process user inputs and display the loan prediction result quickly.	Prediction response within 2–3 seconds
2	Scalability	The application should support multiple users accessing the prediction service simultaneously.	Supports 100+ concurrent users
3	Security & Data Privacy	Applicant information must be securely processed and protected against unauthorized access.	Secure data handling with input validation and controlled access
4	Reliability & Availability	The application should remain available and provide consistent prediction results with minimal downtime.	99% availability during normal operation
5	Usability & Accessibility	The application should provide a simple, responsive, and user-friendly interface that is accessible across modern web browsers.	Responsive UI compatible with Chrome, Edge, Firefox, and mobile browsers
6	Other	The system should be maintainable, modular, and easy to update with new datasets or retrained machine learning models.	Modular architecture supporting future enhancements